

by GParted presently: btrfs, crypt-luks, ext2, ext3, ext4, fat16, fat32, hfs, hfs+.jfs, lvm2 pv, ntfs, reiser4, reiserfs, swap, ufs and xfs. Check GParted.sourceforge.net/features.php for individual support of each filesystem and for updates.

Disk cloning

g4u is a disk cloning software. It is helpful in larger organisations like educational institutes where there are several machines of identical configuration. Every system may need multiple operating systems along with some updates and customisation. Repeating these for every system is a tedious task. So cloning is a good technique to avoid such problems by copying many disks from one fully prepared one. Also, when one or more operating systems crash on one system, it can be recovered quickly by copying the OS from an identical system.

This software has a few advantages over other cloning utilities. One is that it's free. Second, it is independent of partitions, filesystems and operating systems that exist on a disk, as cloning is performed block by block. The only requirement is that the target disk should be larger than the original disk.

g4u comes as live media, in the form of two bootable floppies or a bootable CD. The images for floppies/CD can be downloaded from www.feyrer.de/g4u/#reqs. As of today, the latest stable version of g4u is 2.3.

Disk cloning can be done in two ways — locally, by connecting two or more disks to a system at a time, and remotely, where the image of the original disk is stored on an FTP server, which is used for other disks to be copied using FTP.

So, to perform disk cloning, connect two or more disks for the local copy, or one disk and the network set up if you plan to use FTP and run g4u from a live CD or bootable floppies. A terminal will be presented to use the following commands:

1. **disks** lists all connected disks, and specifies identifiers for each. IDE hard disks are identified by wd0, wd1, etc, whereas SATA disks are identified by sd0, sd1, etc. Here, wd0/sd0 represents the first/primary hard disk.
2. **parts <disk-name>** lists the available partitions in a specified disk. For example, **parts wd0** lists partitions of disks identified by wd0. See Figure 3.
3. **copydisk <disk1> <disk2>** copies all data from disk1 to disk2, block wise (Figure 4). For example, **copydisk wd0 wd1**.
4. **copypart <d1p1> <d2p2>** copies a partition from one disk to a partition on another disk. For example, **copypart wd0e wd1f**, copies the second partition of disk1 to the third partition of disk2. (Here the letters d, e, f, etc, in no way concern the drive letters of any OS. g4u assigns a character for each partition, starting from d for the first partition.)
5. **uploaddisk** stores the image of the whole disk on an FTP server:

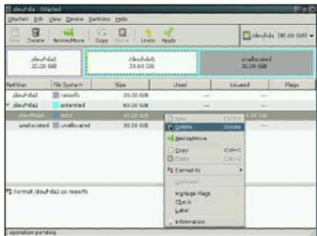


Figure 1: A view of GParted



Figure 2: Resize and move a partition

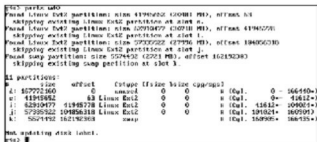


Figure 3: Output of the parts command

`uploaddisk ftp-address imagename disk-id`

...where **ftp-address** is the IP address of the FTP server. By default, the FTP account named 'install' is used. Use **account@ftp-address** for any other account. **disk-id** is the identifier of the hard disk (for example, wd0, wd1, sd0, sd1, etc) whose image needs to be uploaded. It is optional. The absence of **disk-id** implies taking the image of the first hard disk, which is identified by **rw0d0**. Also, by default, **gzip-9** compression is used while saving images with a typical extension like **.gz**. To use lower levels of gzip compressions (higher levels of compression take a longer time to back up/restore) use the following:

`GZIP=1 uploaddisk ftp-address imagename disk-id`

6. **uploadpart** stores the image of a particular partition on the FTP server:

`uploadpart ftp-address imagename disk-id-part`